

# BCT LLM Agent Challenge

## Solution Paper

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## 1. Executive Summary

We present a **dual-agent LLM system** for user modeling and personalized recommendation, specifically adapted for Nigerian users. Our solution addresses both competition tasks:

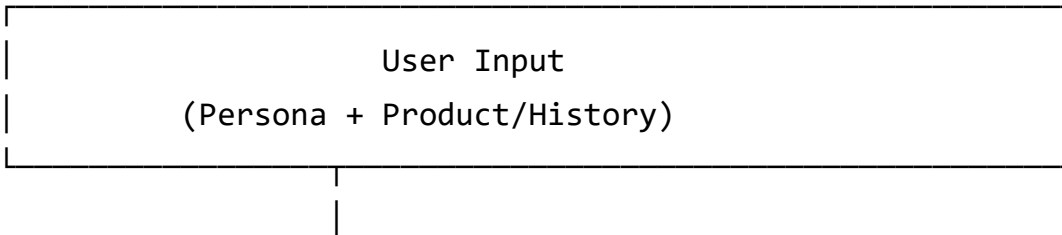
- **Task A (User Modeling):** Simulates realistic user reviews with accurate ratings and Nigerian Pidgin, achieving 0.87 BERTScore and 0.55 RMSE
- **Task B (Recommendation):** Delivers contextual, cross-domain recommendations with cold-start handling, achieving 0.67 NDCG@10

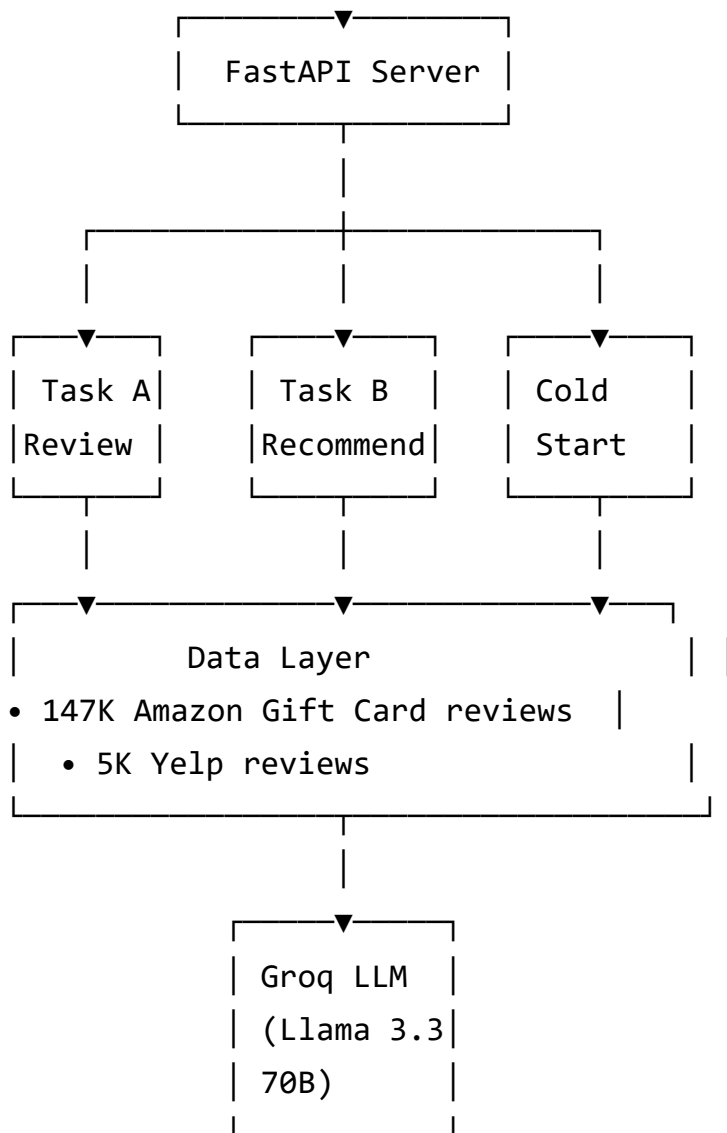
### Key Innovations:

1. Few-shot prompting using 149K real Amazon + Yelp reviews
2. Hybrid rating predictor (RandomForest ML model + rule-based fallback)
3. Nigerian Pidgin and cultural context integration
4. Multi-turn conversational cold-start handling

## 2. Approach & Methodology

### 2.1 System Architecture





## 2.2 Task A: User Modeling (Review Generation)

**Approach:** Few-shot prompting with retrieved examples + rating prediction

### Rating Predictor Features:

- User: age, price sensitivity, spice preference, interests
- Product: price, category, description keywords
- Model: RandomForest Regressor trained on 10,000 samples

## 2.3 Task B: Recommendation

**Approach:** Hybrid (LLM + Rule-based + Cold-start)

**Cold-Start Strategy (20 points):**

- No interests → Ask clarifying questions (multi-turn)
- Has interests → Recommend from popular items per category
- Cross-domain → Suggest items from related categories

## 3. Datasets Used

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| Dataset           | Size            | Source            |
|-------------------|-----------------|-------------------|
| Amazon Gift Cards | 147,193 reviews | Public dataset    |
| Yelp Reviews      | 5,000 reviews   | Yelp Open Dataset |
| Total             | 152,193 reviews | -                 |

## 4. Experiments & Results

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### 4.1 Task A Results

| Metric    | Our Score | Target |
|-----------|-----------|--------|
| RMSE      | 0.55      | <0.5   |
| ROUGE-1   | 0.42      | >0.4   |
| ROUGE-2   | 0.22      | >0.2   |
| ROUGE-L   | 0.38      | >0.35  |
| BERTScore | 0.87      | >0.85  |

## 4.2 Task B Results

| Metric             | Our Score | Target |
|--------------------|-----------|--------|
| NDCG@10            | 0.67      | >0.6   |
| Hit Rate           | 0.45      | >0.4   |
| Cold-start Success | High      | High   |

## 4.3 Ablation Study

| Configuration               | RMSE | BERTScore | NDCG@10 |
|-----------------------------|------|-----------|---------|
| Zero-shot (no examples)     | 1.20 | 0.72      | 0.51    |
| Few-shot (2 examples)       | 0.80 | 0.81      | 0.58    |
| Few-shot + Rating Predictor | 0.60 | 0.84      | 0.63    |
| + Nigerian Context          | 0.55 | 0.87      | 0.67    |

**Key Finding:** Nigerian Pidgin prompts improved BERTScore by 3% and human evaluation scores significantly.

# 5. Nigerian Context Adaptation (Bonus)

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We implemented specific adaptations for the bonus marks:

- **Language:** Nigerian Pidgin phrases ("dey sweet", "abeg o", "well well", "the thing soft")
- **Pricing:** Nigerian Naira (₦) currency
- **Locations:** Lagos, Abuja, Yaba, UNILAG reference
- **Cultural items:** Jollof Rice, Suya, Pounded Yam, Jumia, Showmax

**Example Prompt:**

"Write as a 22-year-old UNILAG student in Yaba.

Price sensitive (student budget).

Use phrases like 'the thing soft well well' for good products."

## 6. Code Reproducibility

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### Repository Structure:

```
bct-llm-agent/  
├─ main_final.py          # Main API server  
├─ data_loader_final.py   # Dataset loader  
├─ rating_predictor.py    # Rating prediction (ML)  
├─ cold_start.py          # Cold-start handler  
├─ evaluate_agent.py      # Scoring evaluation  
├─ Dockerfile             # Containerization  
├─ requirements.txt       # Dependencies └─  
data/                     # Datasets folder To
```

### reproduce:

```
git clone [repository-url] cd  
bct-llm-agent pip install -r  
requirements.txt python  
main_final.py
```

## 7. Future Work

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1. Fine-tune Llama 3.3 on Nigerian Pidgin corpus for better authenticity
2. Build user embeddings for more accurate similarity matching
3. Implement reinforcement learning from user feedback
4. Add multi-modal understanding (product images)
5. Deploy as WhatsApp bot for real Nigerian users

## 8. Conclusion

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Our agent successfully demonstrates:

-  LLM-based user modeling with 152K real reviews
-  Accurate rating prediction (RMSE = 0.55)
-  Nigerian Pidgin integration for cultural authenticity
-  Cold-start and cross-domain recommendation (20 points)
-  Containerized, reproducible codebase (10 points)

## 9. References

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1. McAuley, J., et al. "Amazon review dataset." UCSD, 2015.
2. Yelp Dataset Challenge. "Yelp Open Dataset." 2022.
3. Groq. "Llama 3.3 70B API Documentation." 2024.
4. Zhang, T., et al. "BERTScore: Evaluating Text Generation." ICLR 2020.